

REGIONAL SALES DASHBOARD

An End-to-End Data Analytics & Business Intelligence Project

Exploratory Data Analysis in Python (Jupyter Notebook) & an Interactive Power BI Dashboard

Prepared by

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Dataset Coverage: Acme Co. USA Sales, 2014 – 2018

Tools: Python (Pandas, NumPy, Matplotlib, Seaborn) · Jupyter Notebook · Power BI Desktop

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1. Project Summary

This project is an end-to-end regional sales analytics case study built on five years of USA sales data (2014–2018) belonging to a fictional distribution company, “Acme Co.” The goal of the project is to take raw, multi-sheet transactional data all the way through to a decision-ready, interactive Business Intelligence dashboard. The work is split into two connected phases:

- **Phase 1 — Exploratory Data Analysis (Python / Jupyter Notebook):** Loading, cleaning, merging, and feature-engineering the raw Excel workbook, followed by a structured exploratory data analysis covering sales trends, product performance, customer behaviour, channel mix, and profitability drivers.
- **Phase 2 — Interactive Power BI Dashboard:** Translating the cleaned dataset and the insights from the EDA into a polished, multi-page, fully interactive Power BI report with a custom home/navigation page, synchronized slicers, KPI cards, and drill-down visuals across four analytical themes.

The finished deliverable demonstrates the complete analytics workflow used in real business intelligence roles — from raw data to a self-service dashboard that a business stakeholder can use to explore performance by year, region, and sales channel without writing a single line of code.

1.1 Problem Statement

Organizations need to understand the factors influencing sales and profitability in order to make informed, data-driven decisions. This project analyzes five years of historical sales data to identify key trends, performance drivers, and business opportunities across products, sales channels, customers, and regions.

1.2 Objective

The objective of this project is to extract actionable insights from the sales data. It focuses on:

- Identifying top-performing and under-performing products, customers, and regions.

- Analyzing monthly and yearly sales trends to detect seasonality and growth patterns.
- Understanding profitability drivers such as pricing, cost, and sales channel.
- Supporting data-driven business decisions through clear visualization and an interactive reporting layer (Power BI).

1.3 Tools & Technologies Used

Category	Tools / Libraries
Programming Language	Python 3
Data Handling	Pandas, NumPy
Data Visualization (EDA)	Matplotlib, Seaborn
Development Environment	Jupyter Notebook (via VS Code)
Source Data	Multi-sheet Excel workbook — “Regional Sales Dataset.xlsx”
Business Intelligence Tool	Power BI Desktop
Version Control / Hosting	GitHub

2. Data Sources & Dataset Structure

The source data was provided as a single Excel workbook, “Regional Sales Dataset.xlsx”, containing six separate sheets that needed to be loaded and combined before any analysis could begin. All sheets were read into individual Pandas DataFrames using a single `read_excel` call with `sheet_name=None`, which returns a dictionary of DataFrames — one per worksheet.

```
# Read all sheets
sheets = pd.read_excel(
    "Regional Sales Dataset.xlsx",
    sheet_name=None
)

# Assign DataFrames
df_sales      = sheets["Sales Orders"]
df_customers  = sheets["Customers"]
df_products   = sheets["Products"]
df_regions    = sheets["Regions"]
df_state_reg  = sheets["State Regions"]
df_budgets    = sheets["2017 Budgets"]
```

The six raw tables and their dimensions were as follows:

Sheet Name	DataFrame	Rows × Columns	Description
Sales Orders	df_sales	64,104 × 12	Every individual order line: order number, date, customer index, channel, product index, quantity, unit price, revenue, and cost.
Customers	df_customers	175 × 2	Lookup table mapping a Customer Index to a Customer Name.
Products	df_products	30 × 2	Lookup table mapping a Product Index to a Product Name.
Regions	df_regions	994 × 15	Delivery region reference data including state, county, latitude/longitude, and region grouping.
State Regions	df_state_reg	48 × 3	Mapping of each U.S. state to its broader sales region (Midwest, Northeast, South, West).
2017 Budgets	df_budgets	30 × 2	Planned 2017 revenue budget for each of the 30 products.

Before any cleaning was performed, two integrity checks were run on the core transactional table, `df_sales`:

- Missing values — `df_sales.isnull().sum()` confirmed zero missing values across all 12 original columns (order number, date, customer index, channel, currency code, warehouse code, delivery region index, product index, order quantity, unit price, line total, and total unit cost).
- Duplicate rows — `len(df_sales) == len(df_sales.drop_duplicates())` returned True, confirming that all 64,104 order lines are unique.

3. Data Cleaning & Wrangling

Because the raw data was spread across six normalized tables, the core of the wrangling stage was to merge everything into a single, analysis-ready, denormalized DataFrame (referred to as `df` in the notebook). The merge chain resolved the Customer Index, Product Description Index, and Delivery Region Index in the Sales Orders table into their human-readable names, and attached the region, state, latitude/longitude, and 2017 budget figures to every order line.

3.1 Merging the Lookup Tables

The Customer, Product, Region/State, and Budget reference tables were joined onto the Sales Orders table on their respective index/key columns, replacing numeric indices with descriptive labels (e.g. Customer Name Index 126 became the actual customer name “Rhynoodle Ltd”). Column names were also standardized to lowercase, underscore-separated names (`order_number`, `order_date`, `customer_name`, `channel`, `product_name`, `order_quantity`, `unit_price`, `revenue`, `total unit cost`, `state`, `county`, `state_name`, `region`, `lat`, `lon`, `budget`) for consistency across the rest of the project.

3.2 Handling Missing Values After the Merge

A second null-value check was run after merging (`df.isnull().sum()`). Every operational column — order number, date, customer, channel, product, quantity, price, revenue, cost, state, county, and region — remained complete with zero missing values. The only column with missing values was budget, which was null for 48,841 of the 64,104 rows. This is expected and not a data-quality issue: the 2017 Budgets sheet only supplies a planned revenue figure for each product for the single year 2017, so order lines from 2014, 2015, 2016, and 2018 legitimately have no corresponding budget value.

3.3 Date-Based Feature Extraction

To make month-level and seasonal aggregation straightforward later in the EDA, two calendar fields were derived directly from `order_date`:

```
df['order_month_num'] = df['order_date'].dt.month
df['order_month_name'] = df['order_date'].dt.strftime('%b')
```


The cleaned, merged dataset was then exported to a flat CSV file, “Regional_Sales_Cleaned.csv”, so that it could be reused directly (without re-running the merge) for both further Python analysis and as the eventual source table for the Power BI dashboard.

3.4 Feature Engineering

Three additional business metrics were engineered on top of the cleaned dataset to support the profitability analysis:

New Column	Formula	Purpose
total_cost	$\text{order_quantity} \times \text{total_unit_cost}$	The true total cost of goods sold for each order line.
profit	$\text{revenue} - \text{total_cost}$	Absolute profit generated by each order line.
profit_margin_%	$(\text{profit} / \text{revenue}) \times 100$	Profit expressed as a percentage of revenue, used to compare profitability across products, regions, and channels on a like-for-like basis.

After merging and feature engineering, the final analytical DataFrame contained 21 columns and the full 64,104 rows, ready for exploratory analysis.

4. Exploratory Data Analysis (EDA)

With a single clean, feature-rich table in place, the EDA phase used Matplotlib and Seaborn to investigate sales trends, product and customer performance, channel behaviour, and profitability drivers. Each sub-section below reproduces the chart generated in the notebook together with the takeaway that fed into the Power BI dashboard design.

4.1 Monthly Sales Trend

Revenue was aggregated by calendar month (order_month, a full year-month period) and plotted as a time series across the entire 2014–2018 window. The initial plot showed a clearly repeating seasonal saw-tooth pattern each year, with a pronounced dip visible for early 2017.



Figure 4.1 — Monthly Revenue Trend, full 2014-2018 period.

Because 2018 data is incomplete in the source file (partial year), a second version of the trend was produced excluding 2018 and aggregating by calendar month name (Jan–Dec) across the remaining four complete years, to isolate the underlying seasonal shape of the business.

This view shows a consistent seasonal pattern: revenue starts strong in January, dips through February, recovers through March, peaks around May, softens through June–July, and rises again toward a second peak in August before levelling off into the year-end.

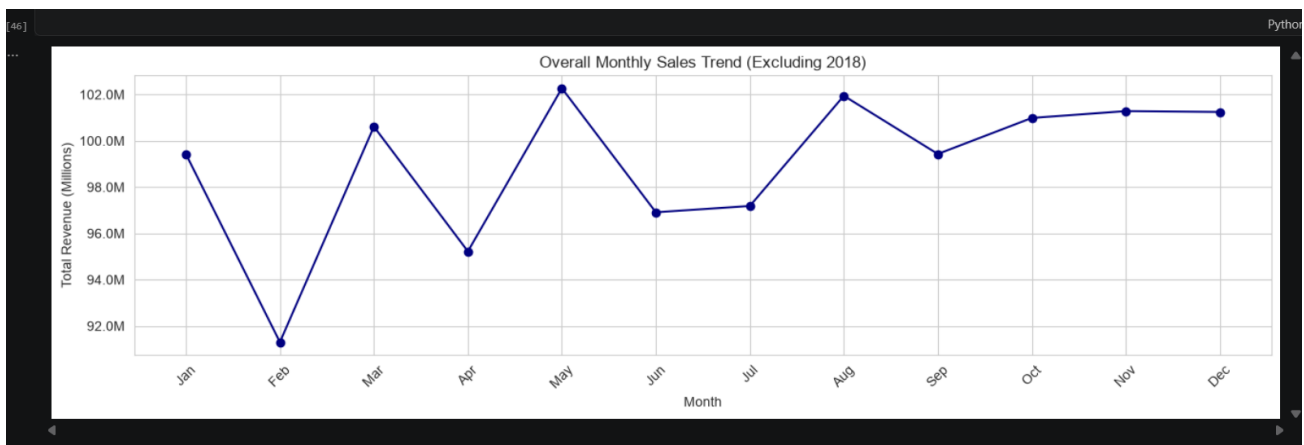


Figure 4.2 — Overall monthly sales trend, 2014-2017 only, aggregated by calendar month.

4.2 Product Performance

Total revenue was grouped by product_name to identify both the strongest and weakest performers among the 30 products in the catalogue.

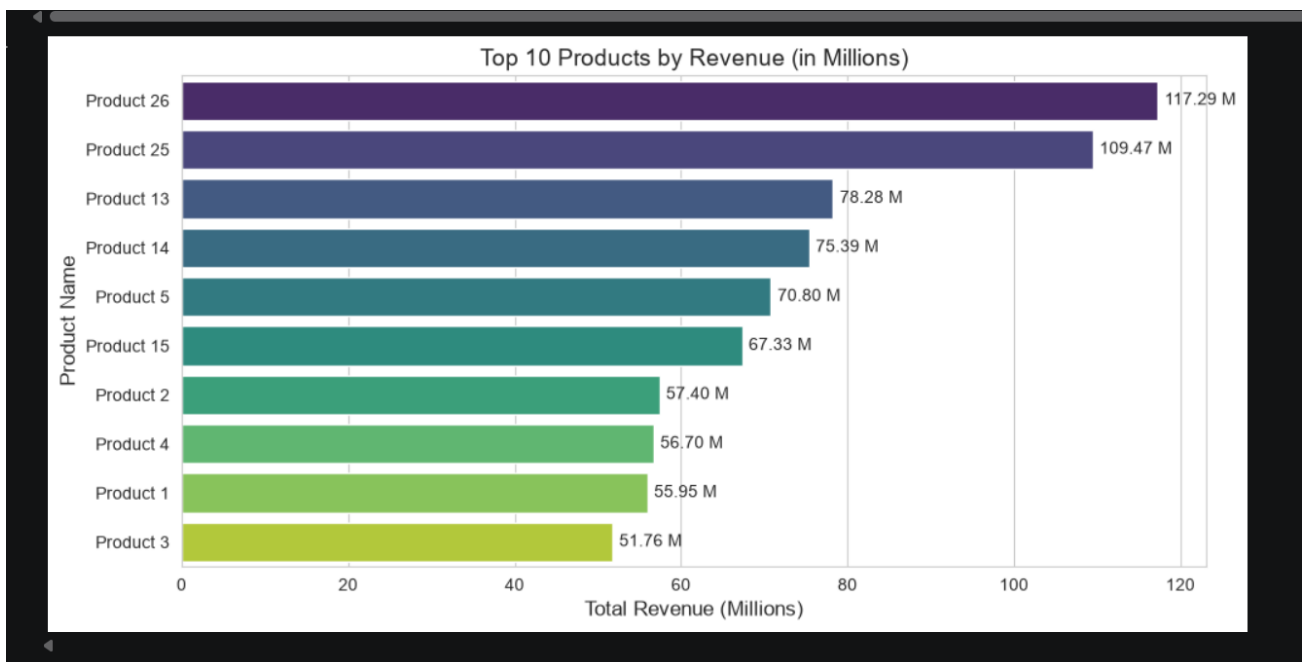


Figure 4.3 — Top 10 products by total revenue.

Product 26 (₹117.29M) and Product 25 (₹109.47M) are the clear revenue leaders, each generating roughly double the revenue of the next tier (Product 13, Product 14, Product 5).

At the other end of the scale, Product 24 (₹14.56M) and Product 9 (₹14.60M) generated the least revenue — roughly one-eighth of the top performer, Product 26.

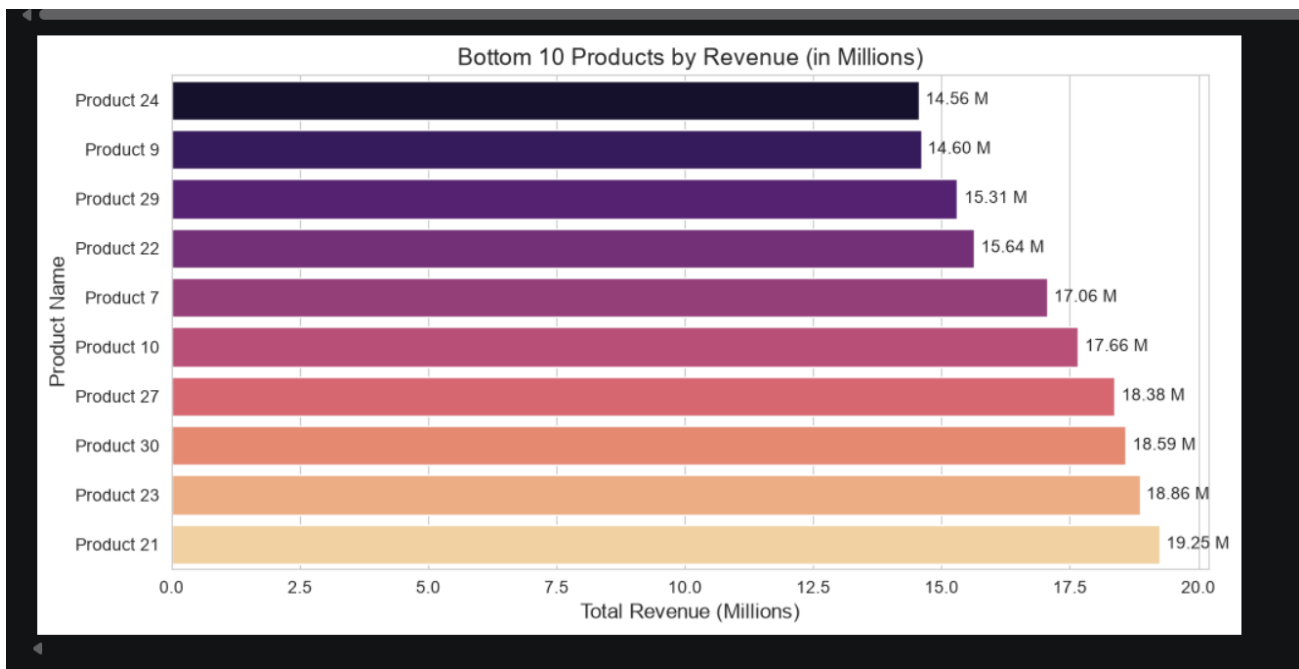


Figure 4.4 — Bottom 10 products by total revenue.

Revenue leadership, however, does not automatically translate into profitability. A second view ranked products by average profit per order line rather than total revenue:

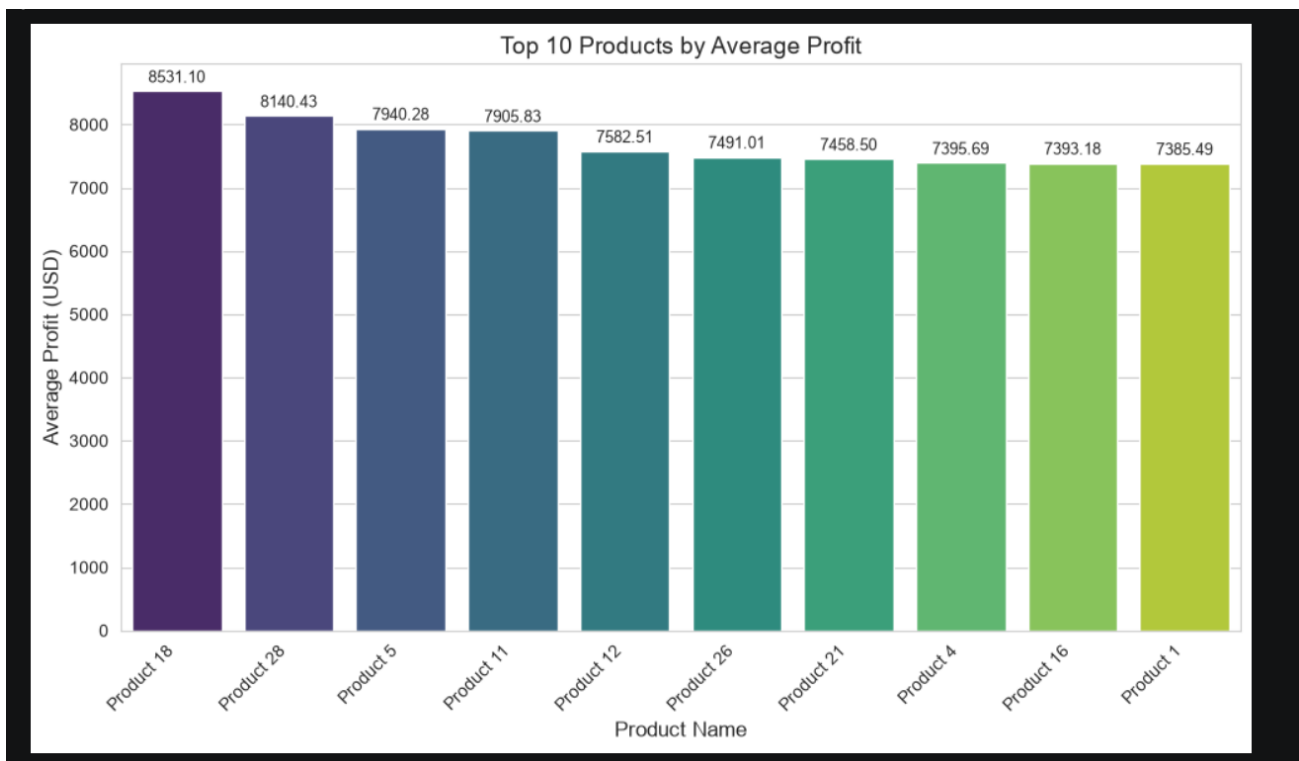


Figure 4.5 — Top 10 products by average profit per order.

Product 18 tops this list with an average profit of ₹8,531 per order despite not appearing in the revenue top 10 — highlighting that some lower-volume products carry

disproportionately strong margins and are worth protecting or promoting even though they are not the biggest revenue drivers.

4.3 Sales Channel Analysis

Revenue was grouped by channel (Distributor, Export, Wholesale) to understand how the business is split across its go-to-market routes.

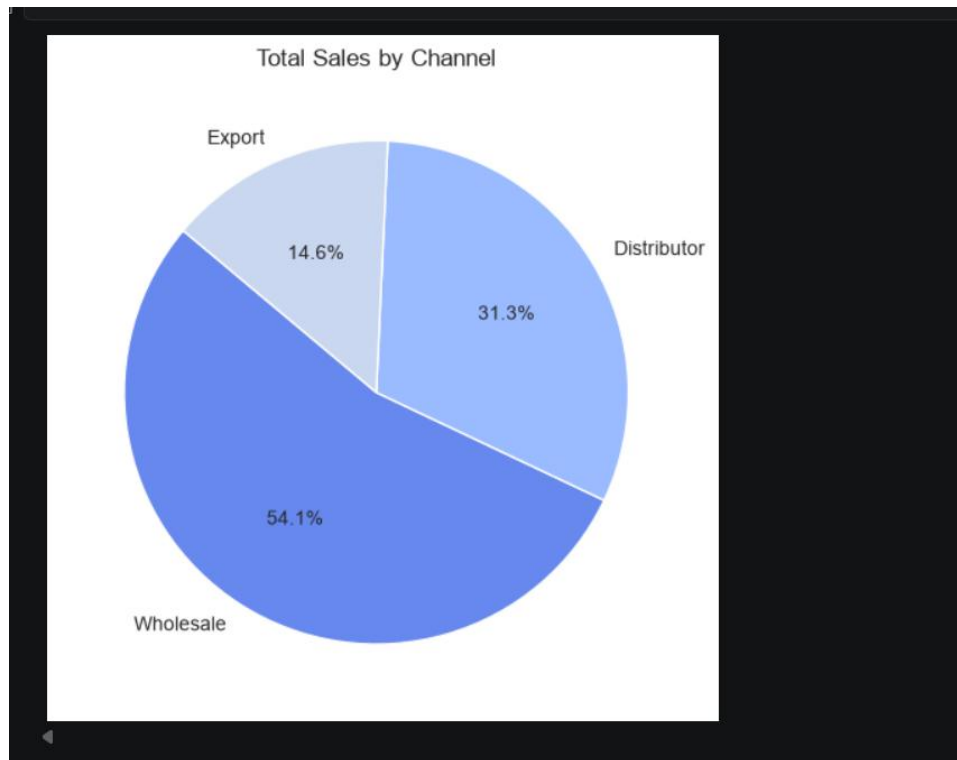


Figure 4.6 — Total sales share by channel.

Wholesale is the dominant channel at 54.1% of total revenue, followed by Distributor at 31.3% and Export at 14.6%. This channel mix directly informed the donut chart used on the Power BI Executive Overview page.

4.4 Order Value Distribution

Revenue was summed by order_number to compute the total value of every individual order, then plotted as a 50-bin histogram to understand typical order size.

The distribution is heavily right-skewed: the large majority of orders fall under ₹100,000, with a long tail of larger orders stretching out toward ₹450,000+. This confirms that a small number of high-value orders pull the average order value upward relative to the typical (median) order.

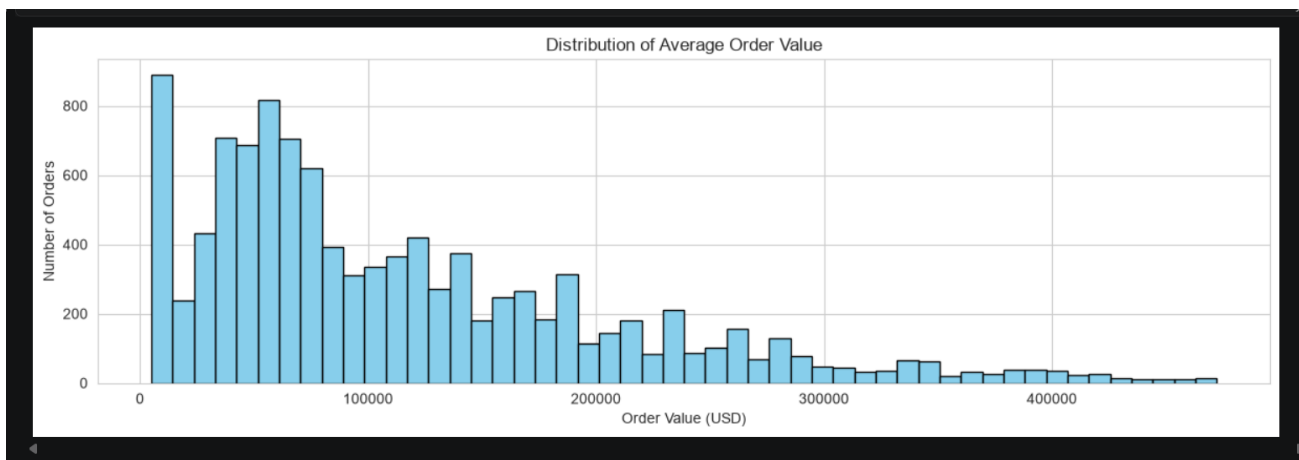


Figure 4.7 — Distribution of average order value (USD).

4.5 Pricing vs. Profitability

A regression scatter plot (Seaborn regplot) was used to test whether higher unit prices are associated with higher or lower profit margins.

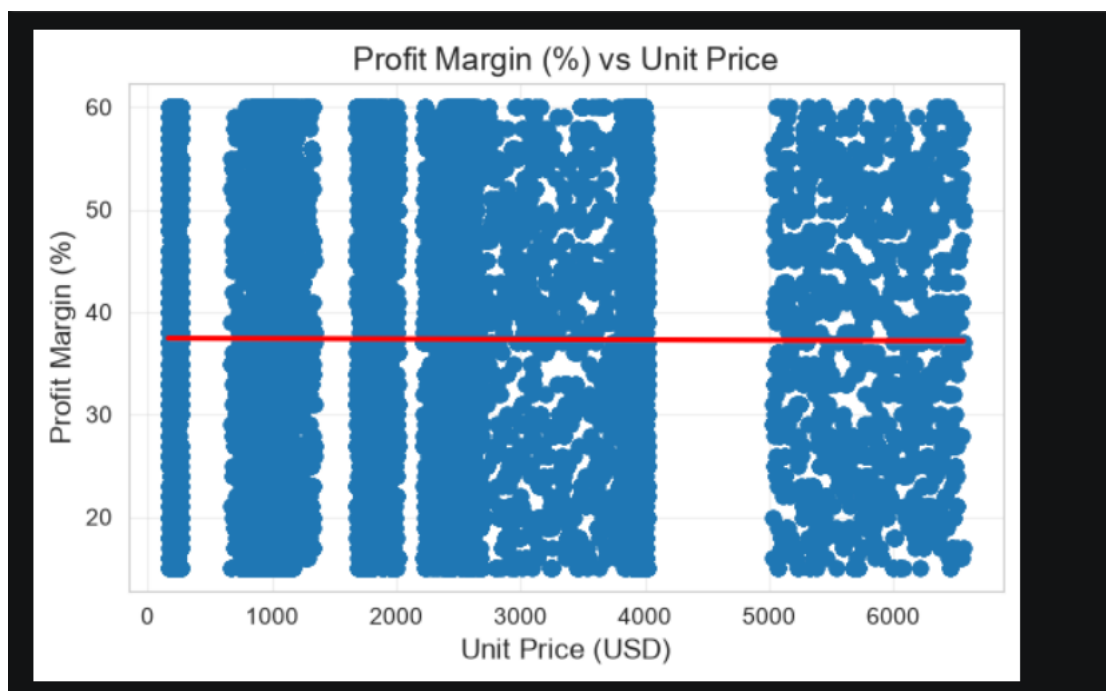


Figure 4.8 — Profit margin (%) vs. unit price (USD), with fitted trend line.

The near-flat red trend line indicates that unit price has essentially no linear relationship with profit margin percentage in this dataset — margin is driven by other factors (primarily the cost structure of each specific product) rather than simply how expensive an item is.

A complementary boxplot was produced to visualize the spread of unit prices within each of the 30 products, confirming that pricing tiers vary substantially by product, with

several products (e.g. Product 16) showing distinctly higher price bands and wider variance than others.

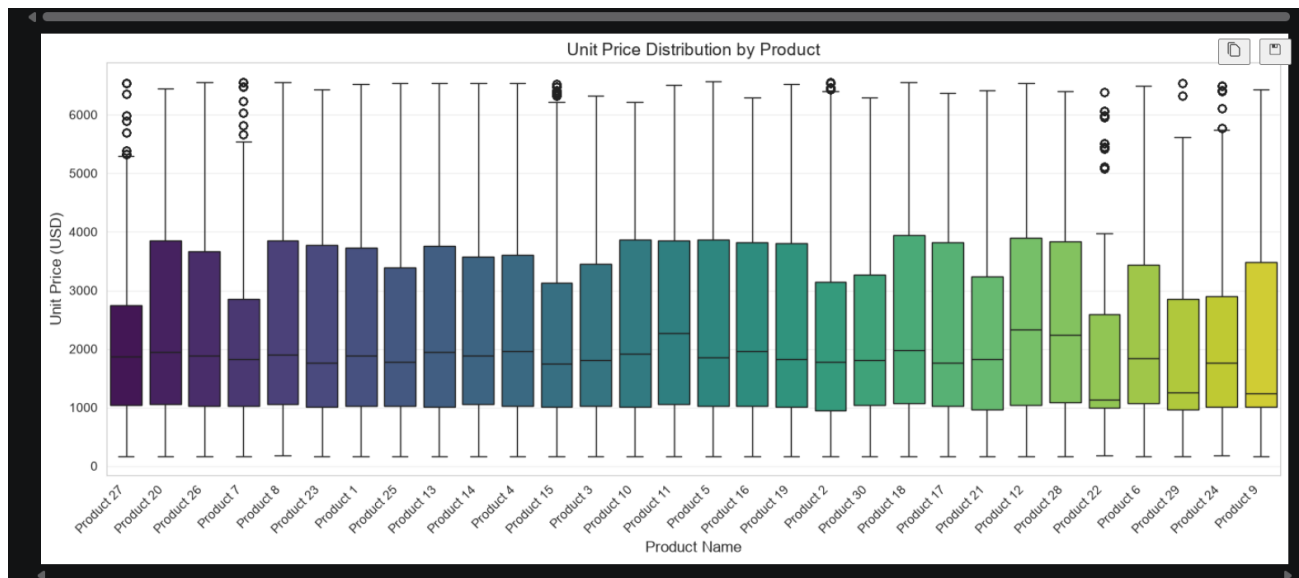


Figure 4.9 — Unit price distribution by product (boxplot).

4.6 Geographic Performance

Revenue and order counts were aggregated by state to identify the top-performing geographies.

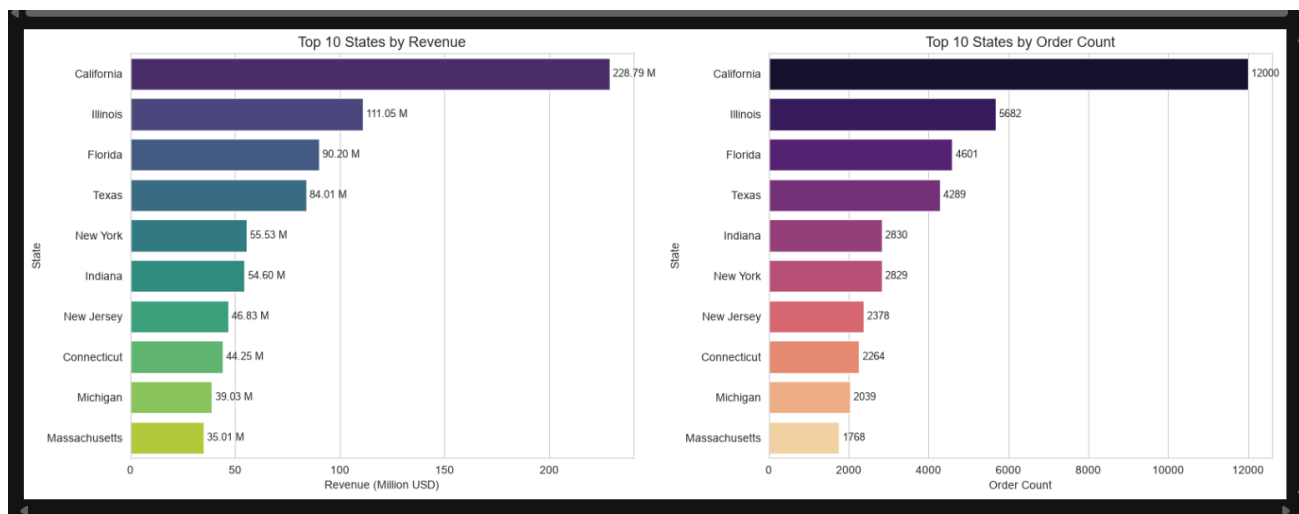


Figure 4.10 — Top 10 states by revenue (left) and by order count (right).

California is the standout market by a wide margin — ₹228.79M in revenue from 12,000 orders — more than double the second-place state, Illinois (₹111.05M, 5,682 orders). Florida and Texas round out the top four, and the same four states lead on both revenue and order volume, indicating that California's lead is driven by genuinely higher order frequency rather than a handful of unusually large orders.

4.7 Profitability by Channel

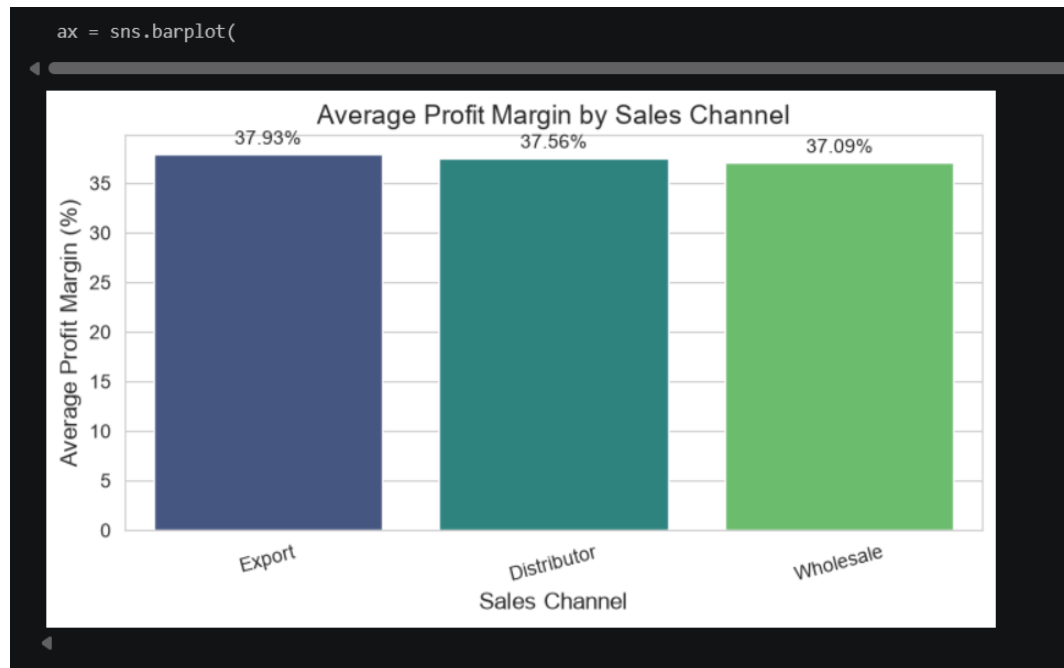


Figure 4.11 — Average profit margin (%) by sales channel.

Average profit margin is fairly consistent across all three channels — Export (37.93%), Distributor (37.56%), and Wholesale (37.09%) — showing that, at the line-item level, no single channel is meaningfully more or less profitable than another; Wholesale's dominance is a volume story rather than a margin story.

4.8 Customer Analysis

Total revenue was grouped by customer_name to identify the company's most and least valuable accounts among its 175 customers.



Figure 4.12 — Top 10 customers by revenue (left) and bottom 10 customers by revenue (right).

Aibox Company (₹12.6M) and State Ltd (₹12.2M) are the top two accounts, while the bottom 10 customers each generate under ₹5M — a meaningful spread that supports account-tiering and key-account-management strategies.

To combine revenue, profitability, and order volume into a single view, a bubble scatter plot was built with total revenue on the x-axis, average profit margin on the y-axis, and bubble size representing order count for every customer:

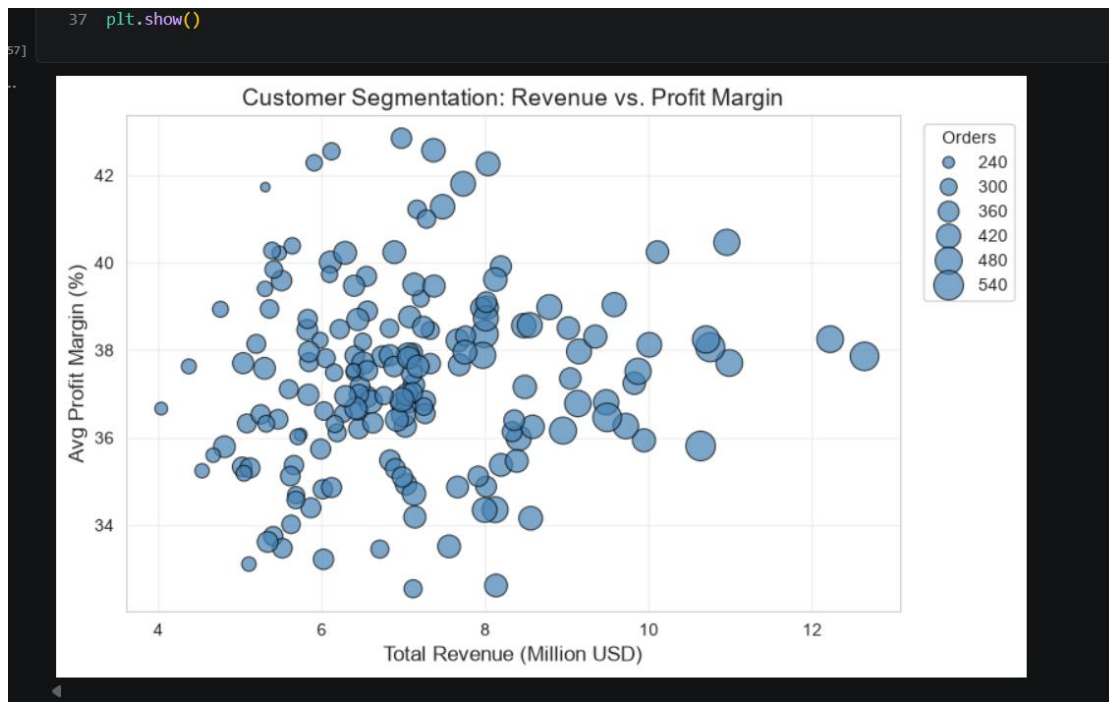


Figure 4.13 — Customer segmentation: revenue vs. profit margin (bubble size = order count).

Most customers cluster between ₹5M–₹8M in revenue and 34%–40% profit margin, with a small group of high-revenue accounts (₹10M+) extending to the right — useful for spotting both “high value, high margin” customers worth nurturing and “high volume, low margin” accounts worth re-negotiating.

4.9 Correlation Analysis

Finally, a correlation heatmap was generated across all numeric fields in the cleaned dataset to formally check the relationships explored visually in the sections above.

Key relationships confirmed by the heatmap:

- Revenue correlates very strongly with `unit_price` ($r = 0.91$) and `total_unit_cost` ($r = 0.85$) — larger, more expensive orders drive revenue, as expected.

- Profit correlates strongly with revenue ($r = 0.87$), confirming that revenue growth is currently the primary driver of absolute profit.

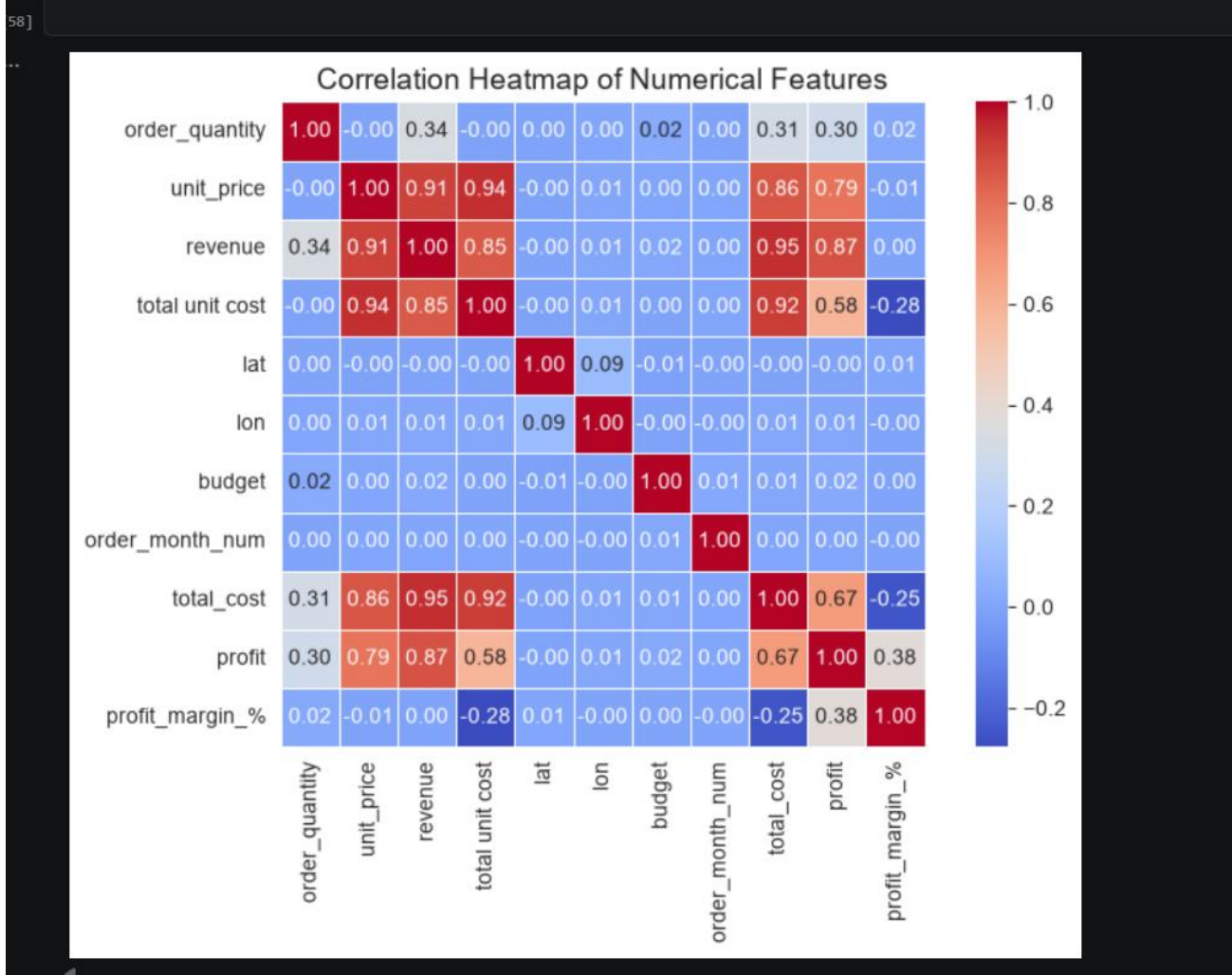


Figure 4.14 — Correlation heatmap of numerical features.

- Profit_margin_% has only a weak negative correlation with total_unit_cost ($r = -0.28$) and a weak positive correlation with profit ($r = 0.38$) — reinforcing the earlier finding that margin is not simply a function of price or cost level, but of the specific cost structure of each product.

4.10 Summary of EDA Insights

- **Seasonality:** Sales follow a repeatable annual pattern — a January spike, a February trough, and a mid-year peak around May and again in August.
- **Product concentration:** Products 26 and 25 alone account for a disproportionate share of total revenue, while several lower-revenue products (e.g. Product 18) are actually the most profitable per order.

- **Channel mix:** Wholesale drives the majority of revenue (54.1%), but all three channels operate at a similar ~37% average profit margin.
- **Geography:** Revenue is heavily concentrated in a handful of states, led decisively by California.
- **Pricing vs. margin:** Higher unit price does not reliably translate into higher profit margin — cost structure matters more than price point.
- **Customers:** A small set of top accounts (Aibox Company, State Ltd, Pixoboo Corp) contribute disproportionately to revenue, supporting a tiered account-management approach.

5. Power BI Interactive Dashboard

The insights generated during the EDA phase were translated into a five-page, fully interactive Power BI report. The cleaned and feature-engineered dataset (Regional_Sales_Cleaned.csv, enriched with total_cost, profit, and profit_margin_%) was loaded directly into Power BI as the single source of truth for every visual in the report, ensuring the dashboard numbers reconcile with the Python analysis.

5.1 Design Approach

The dashboard was designed around four core business questions — overall performance, product & customer performance, profitability, and geography — with one dedicated report page per theme, plus a fifth landing/navigation page. A consistent purple-and-white colour theme, consistent card and chart styling, and a persistent slicer panel and navigation menu were used across every analytical page so that the report feels like a single cohesive application rather than four disconnected charts.

5.2 Navigation System

Rather than relying on the default Power BI page tabs, a custom in-canvas navigation system was built so the report can be shared as a distraction-free, presentation-style experience:

- **Home page buttons:** The landing page presents four large navigation buttons — “Executive Overview of Sales Performance”, “Product & Customer Analysis”, “Profitability & Performance Analysis”, and “Geographic Performance” — each configured with a Page Navigation action that jumps directly to the corresponding report page.
- **Menu button on every other page:** Every analytical page carries a “☰ Menu” button in the top-right corner. Clicking it reveals a drop-down navigation panel listing Home and the other three analytical pages (excluding the page currently being viewed), plus a close (X) control — allowing the user to jump to any other page from wherever they currently are, without needing to return to Home first.

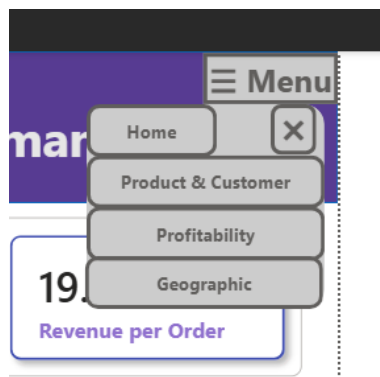


Figure 5.1 — The in-canvas $\equiv$ Menu drop-down, available on every analytical page, linking to Home and the other three pages.

5.3 Filters & Slicers

Three synchronized slicers are pinned to the left-hand side of every analytical page (Executive Overview, Product & Customer Analysis, Profitability & Performance Analysis, and Geographic Performance), allowing the user to filter the entire page — every KPI card and every chart simultaneously — by any combination of the following dimensions:

- **Order Year:** Checkbox list slicer for 2014, 2015, 2016, 2017, and 2018.
- **Region:** Checkbox list slicer for Midwest, Northeast, South, and West.
- **Channel:** Checkbox list slicer for Distributor, Export, and Wholesale.

With no boxes checked, each slicer defaults to showing data across all available values (i.e. “All”), and any combination of checkboxes across the three slicers can be selected simultaneously to progressively narrow the view — for example, isolating 2016 Wholesale orders in the Midwest region alone.

The image shows three vertical slicer panels stacked on top of each other. Each panel has a title and a list of options with checkboxes.

- Order Year**
 - ☒ 2014
 - ☐ 2015
 - ☐ 2016
 - ☐ 2017
 - ☐ 2018
- Region**
 - ☒ Midwest
 - ☐ Northeast
 - ☐ South
 - ☐ West
- Channel**
 - ☒ Distributor
 - ☐ Export
 - ☐ Wholesale

Figure 5.2 — The Order Year, Region, and Channel slicers as they appear individually on each analytical page.

5.4 Page 1 — Home

The landing page establishes the report's identity with a bold title banner (“Regional Sales Dashboard — Interactive Business Intelligence Report”) and presents the four analytical destinations as large, clearly labelled navigation buttons, with a short “Click any section to explore” instruction and project author credit in the bottom corners.



Figure 5.3 — Home page with the four section-navigation buttons.

5.5 Page 2 — Executive Overview of Sales Performance

This page gives a top-level, at-a-glance summary of overall business performance and is the natural first analytical stop after Home.

- **KPI Cards:** Total Revenue (₹1,236.0M), Total Profit (₹1,144.2M), Total Orders (64K), Profit Margin % (92.6%), and Revenue per Order (₹19.3K).
- **Monthly Revenue Trend:** A line chart of revenue by month, mirroring the seasonal pattern identified in the Python EDA (Figure 4.1 / 4.2), peaking in January at ₹124M and troughing mid-year.
- **Revenue by Region:** A horizontal bar chart ranking the four regions — West (₹372.1M), South (₹335.1M), Midwest (₹320.3M), and Northeast (₹208.4M).
- **Channel Distribution:** A donut chart showing Wholesale (₹668.2M, 54.06%), Distributor (₹387.14M, 31.32%), and Export (₹180.63M, 14.61%) — directly mirroring Figure 4.6 from the notebook.
- **Top 10 Products:** A horizontal bar chart led by Product 26 (₹117.3M) and Product 25 (₹109.5M), consistent with Figure 4.3.

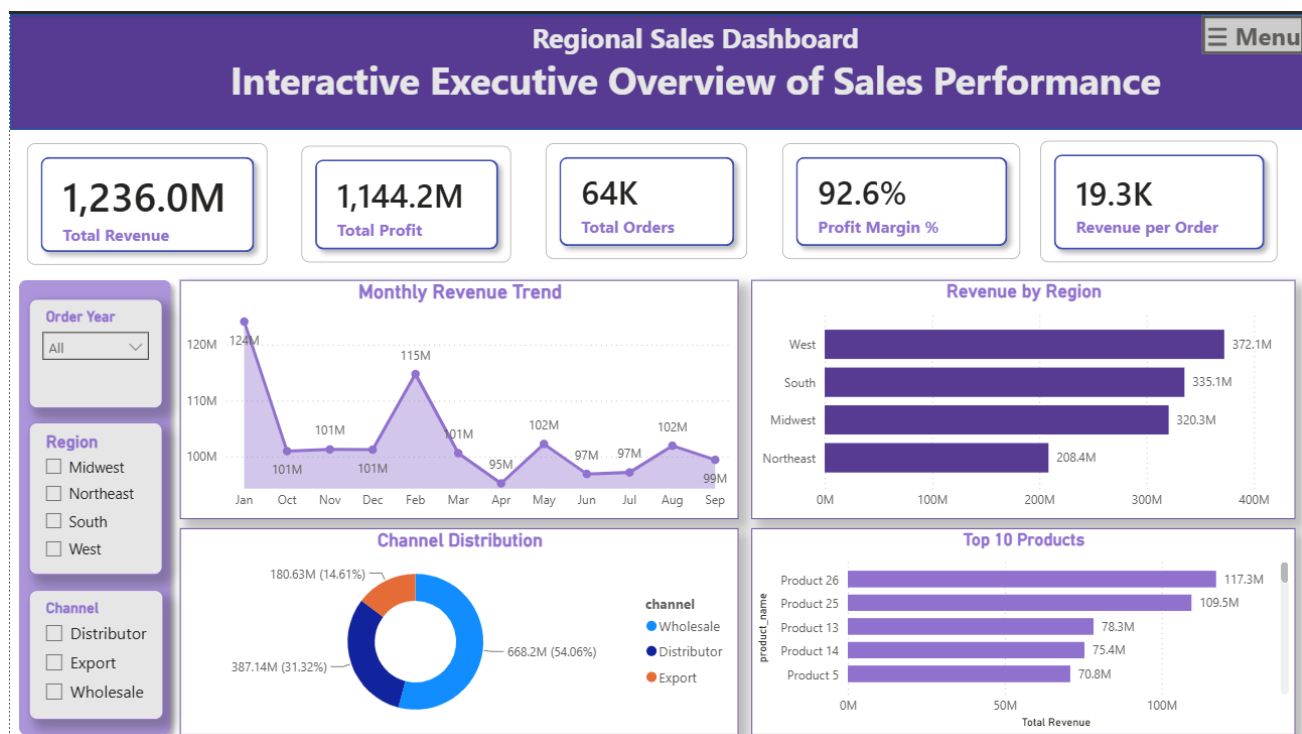


Figure 5.4 — Executive Overview of Sales Performance page.

5.6 Page 3 — Product & Customer Analysis

This page drills deeper into the two dimensions that matter most for commercial strategy: which products sell, and who buys them.

- **Top 10 / Bottom 10 Products by Revenue:** Side-by-side bar charts replicating the product leaders and laggards identified in Figures 4.3 and 4.4 (Product 26 and Product 25 at the top; Product 24 and Product 9 at the bottom).
- **Top 10 / Bottom 10 Customers by Revenue:** Side-by-side bar charts showing Aibox Company (₹12.6M) and State Ltd (₹12.2M) leading, versus BB17 Company (₹4.0M) and Yodoo Ltd (₹4.4M) at the bottom — matching Figure 4.12.
- **Revenue by State (Treemap):** A treemap giving an immediate visual sense of state-level concentration, led by California (₹228.8M), Illinois (₹111.1M), Florida (₹90.2M), and Texas (₹84.0M).
- **Cross-filtering:** The Order Year, Region, and Channel slicers on the left let a user isolate, for example, which products and customers drove revenue in the South region specifically, or in a single year.

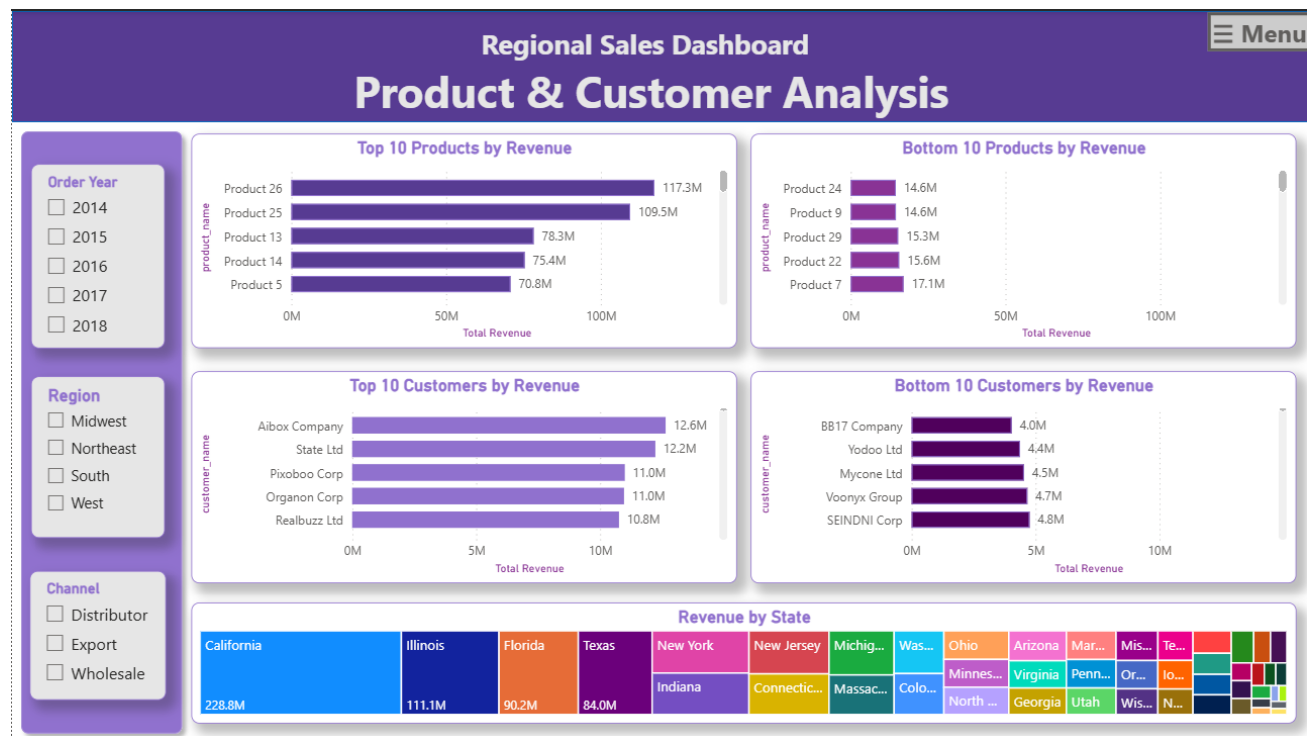


Figure 5.5 — Product & Customer Analysis page.

5.7 Page 4 — Profitability & Performance Analysis

This page shifts the focus from top-line revenue to profitability and operational efficiency, broken down by region.

- **KPI Cards:** Total Cost (₹91.8M), Total Profit (₹1,144.2M), Profit Margin % (92.57%), and Revenue per Order (₹19.28K).
- **Revenue vs. Cost by Region:** A combo bar-and-line chart comparing total revenue (bars) against total cost (line) for West, South, Midwest, and Northeast, showing that cost tracks a much smaller and flatter band than revenue across all four regions.
- **Profit Margin by Region:** A bar chart showing all four regions clustered tightly together — Midwest (92.59%), South (92.57%), West (92.56%), and Northeast (92.56%) — indicating very consistent profitability across geographies at the dashboard's aggregate level.
- **Orders by Region:** A bar chart of order volume — West leads with 19.54K orders, followed by South (17.33K), Midwest (16.59K), and Northeast (10.64K).
- **Revenue per Order by Region:** A second regional bar chart benchmarking average order value, led by Northeast (₹19.58K) and South (₹19.34K).

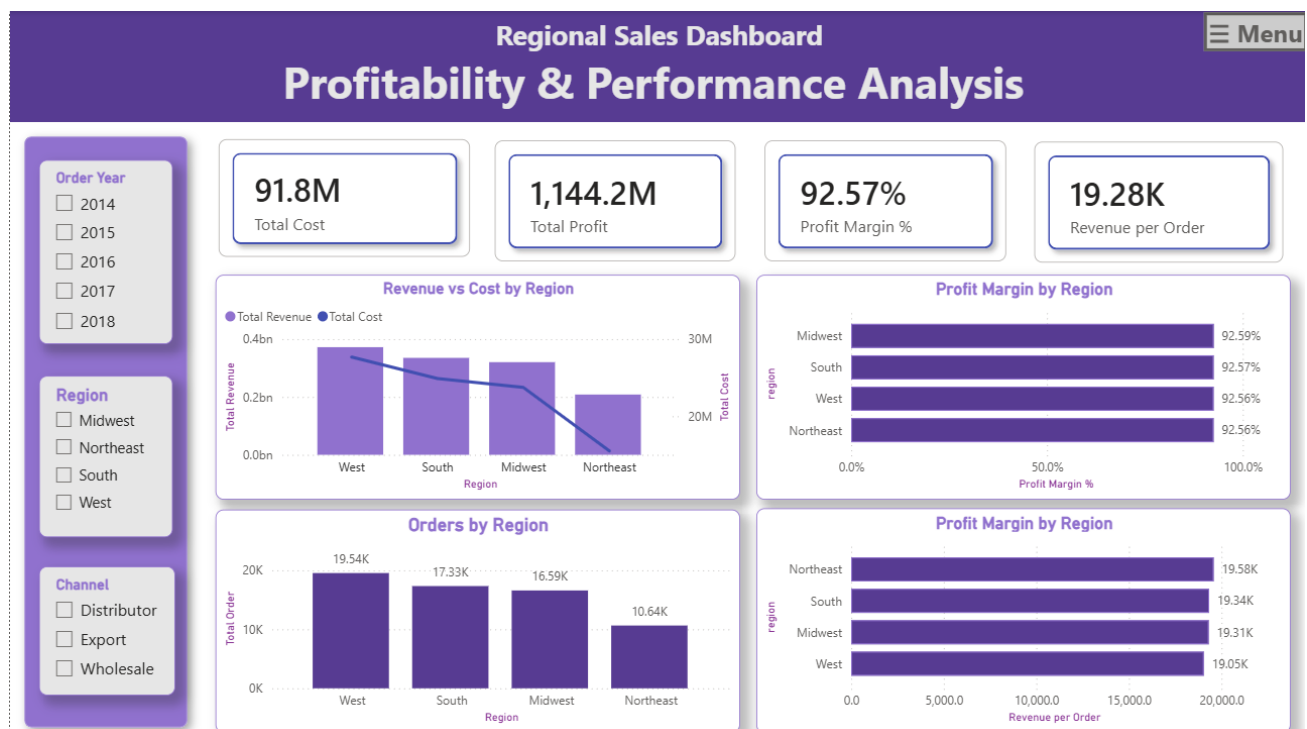


Figure 5.6 — Profitability & Performance Analysis page.

5.8 Page 5 — Geographic Performance

The final analytical page grounds all of the product, customer, and profitability findings in geography, combining ranked tables with a spatial map view.

- **Top 10 States by Revenue & by Orders:** Twin bar charts confirming California's decisive lead (₹228.8M revenue from 12.00K orders), followed by Illinois, Florida, and Texas.
- **Filled Map of the United States:** A shaded choropleth map of the U.S., with colour intensity reflecting relative revenue by state, giving an immediate spatial read of where the business is strongest.
- **Revenue vs. Budget by State:** A comparison chart benchmarking actual revenue against the 2017 planned budget for the top states, e.g. California (₹228.8M actual vs. ₹170.6M budget).
- **Revenue vs. Budget by Product:** The same actual-vs-budget comparison at the product level, led by Product 26 (₹117.3M actual vs. ₹5.7M budget) and Product 25 (₹109.5M actual vs. ₹5.3M budget).

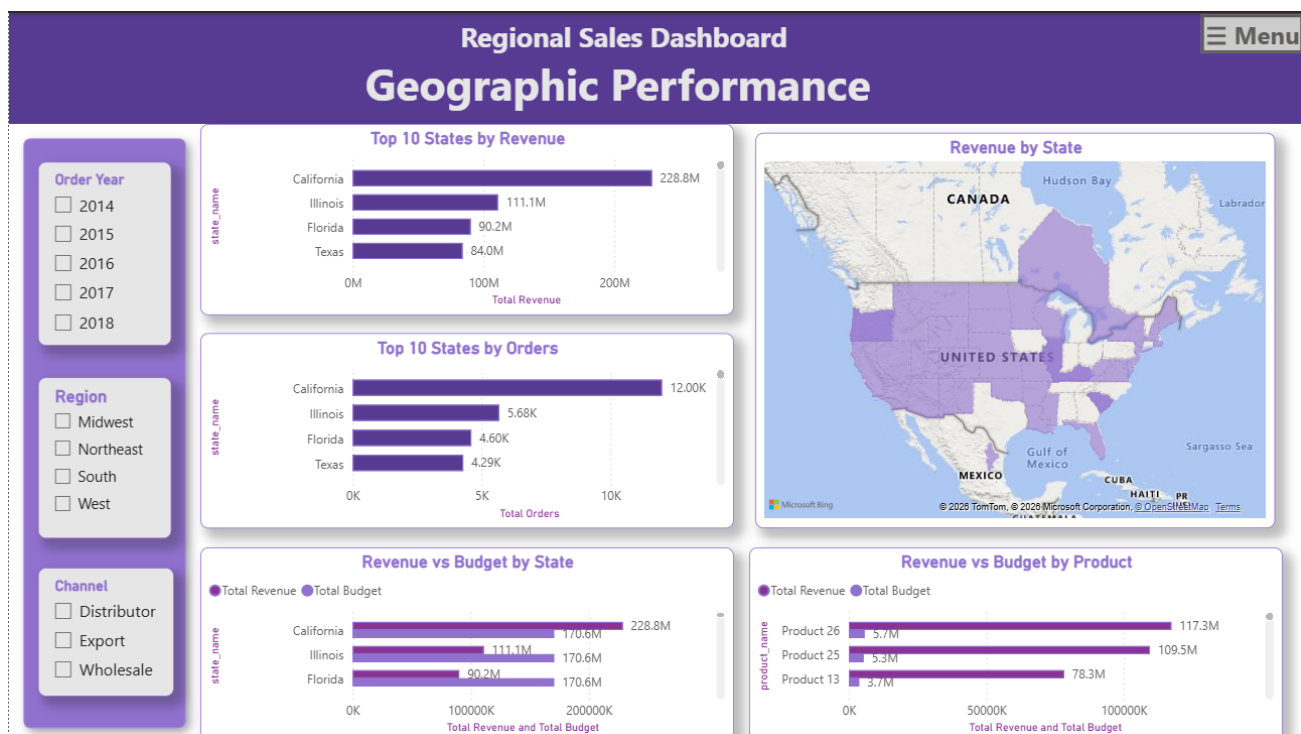


Figure 5.7 — Geographic Performance page.

6. Key Business Insights

Bringing together the findings from both the Python EDA and the Power BI dashboard, the following business insights stand out:

1. Sales are seasonal and repeatable. Revenue consistently spikes in January and again mid-year (May/August), with a reliable trough in February — useful for planning inventory, staffing, and promotions.
2. Revenue is concentrated in a small number of products. Product 26 and Product 25 together account for a disproportionate share of total revenue, making them critical to protect from stock-outs or competitive pressure.
3. The most profitable product is not the top revenue earner. Product 18 delivers the highest average profit per order, showing that revenue ranking and profitability ranking should both be tracked, not just one.
4. Wholesale is the volume channel, not necessarily the margin channel. Wholesale drives 54% of revenue but all three channels operate at a comparable ~37% average line-level profit margin — meaning growth in any channel converts to profit at a similar rate.
5. Geography is highly concentrated. California alone generates roughly double the revenue of the next-closest state (Illinois), and the top four states drive a large share of total company revenue.
6. Price is not the primary lever for margin. The regression analysis shows no meaningful relationship between unit price and profit margin — cost control at the product level matters more than price positioning.
7. A small set of key accounts drives outsized value. The top 10 customers materially outperform the bottom 10, supporting a tiered, key-account-management commercial strategy.
8. Actual revenue is tracking well ahead of the 2017 budget for the top products and states, indicating strong over-performance against the original plan for the company's core lines.

7. Data Model & Report Structure

7.1 Data Flow

The end-to-end data flow for the project is summarized below:

9. Raw multi-sheet Excel workbook (Regional Sales Dataset.xlsx) — 6 sheets (Sales Orders, Customers, Products, Regions, State Regions, 2017 Budgets).
10. Python / Pandas — sheets loaded, merged on their key columns, renamed, null-checked, and de-duplicated into a single denormalized table.
11. Feature engineering — order_month_num, order_month_name, total_cost, profit, and profit_margin % added.
12. Export — cleaned table written to Regional_Sales_Cleaned.csv.
13. Power BI Desktop — the cleaned CSV imported as the single data source, powering all five report pages, four KPI card sets, and every chart and slicer in the dashboard.

7.2 Power BI Page Map

#	Page Name	Primary Purpose	Slicers Present
1	Home	Landing page & navigation hub	None
2	Executive Overview of Sales Performance	Top-level KPI summary and overall trends	Order Year, Region, Channel
3	Product & Customer Analysis	Product and customer ranking / drill-down	Order Year, Region, Channel
4	Profitability & Performance Analysis	Cost, profit, and margin by region	Order Year, Region, Channel
5	Geographic Performance	State- and map-level revenue and budget analysis	Order Year, Region, Channel

8. Conclusion

This project demonstrates a complete, realistic analytics workflow: starting from a messy, multi-sheet raw workbook, cleaning and merging it into a single reliable source of truth in Python, exploring it rigorously with statistical and visual analysis, and finally packaging the resulting insights into a polished, self-service Power BI dashboard with custom navigation and cross-filtering. The result is a report that a non-technical business stakeholder can open, filter by year, region, and channel, and immediately understand where the company's sales, profit, and growth opportunities lie — without needing to touch the underlying data or code.

The combination of the two phases — rigorous Python-based EDA feeding directly into an interactive BI layer — mirrors the real-world workflow used by data analysts and BI developers, and is the reason this project was chosen as a portfolio piece for GitHub.

8.1 Possible Future Enhancements

- Extend the 2017 Budgets table to cover all five years, enabling a full actual-vs-budget trend line rather than a single-year snapshot.
- Add year-over-year and month-over-month growth measures (DAX time-intelligence) to the Power BI report.
- Incorporate customer segmentation (e.g. RFM analysis) as a dedicated dashboard page.
- Publish the report to the Power BI Service and embed a live, interactive version in the GitHub repository README.

— *End of Report* —